

AMARJITH MANJOO

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Education

Indian Institute of Technology (Indian School of Mines), Dhanbad

Jul 2025 – Present

Master of Technology, Fuel and Energy Engineering

CGPA: **9.88 / 10**

Relevant Coursework: Alternate Energy Systems, Biofuels, Coal & Mineral Beneficiation, Fuel Technology, Powerplant Engineering, Computational Fluid Dynamics, Energy Technology, Mineral Economics, Processing of Liquid and Gaseous Fuels

BITS Pilani, Hyderabad Campus

Jul 2020 – May 2024

Bachelor of Engineering, Chemical Engineering

Minor, Finance

CGPA: **7.58 / 10**

Relevant Coursework: Separation Processes, Heat Transfer, Fluid Mechanics, Engineering Chemistry, Kinetics & Reactor Design, Process Design Principles, Process Dynamics And Control, Colloids And Interface Engineering, Environmental Pollution Control

Experience

Master's Thesis exchange student

Sept 2026 - Present

TU Dresden, Germany

- Awarded with the prestigious DAAD KOSPIE Scholarship to do research in Germany for 7 months

Chemical Engineering Intern

May 2022 – Jun 2022

Grasim Industries Ltd, Nagda

- Developed a process simulation in DWSim software to model the Claus-Kiln Plant operations.
- Analyzed sulfur recovery mechanisms to identify operational bottlenecks and proposed strategies to increase overall plant efficiency.

Projects

Concept development and Model based evaluation of Thermochemical Valorization Pathways for biogenic residual streams from tobacco plants

Sept 2026 - Present

TU Dresden, Germany

- Development and quantitative assessment of a thermochemical valorization concept for biogenic residues generated during tobacco processing within the GreenPharming system.

Literature Review of Biohydrogen Production

Jan 2024 – May 2024

- Conducted an extensive review of biohydrogen production technologies, emphasizing recent advancements in sustainable energy.
- Evaluated critical challenges and potential industrial applications for large-scale production.

Graphite Beneficiation Project

Jun 2023 - Dec 2023

- Achieved a 6–8% improvement in graphite recovery by optimizing rod mill charge ratios based on IIT Madras analysis.
- Utilized SEM-EDS, XRD, and TGA characterization techniques to evaluate mineral microstructures and thermal behavior.
- Implemented process optimizations that significantly enhanced the overall efficiency of the beneficiation circuit.

Market Study for Graphene and Spheroidized Graphite

Jul 2023 – Dec 2023

- Researched the Indian market landscape for graphene and SPG to identify key industry players and emerging trends.
- Compiled data-driven insights into supply chain dynamics and market growth potential.

Publications

Advancements in Biohydrogen Production

RSC Advances (Nov 2024)

<https://doi.org/10.1039/D4RA06214K>

- Co-authored a high-impact review evaluating the efficiency of batch, continuous, and membrane reactors, specifically identifying optimal operational ranges for temperature and acidic pH.
- Assessed the environmental and economic viability of production through Life Cycle Assessment (LCA) and Techno-Economic Analysis (TEA) to address scalability challenges and high operating costs.
- Analyzed substrate-specific performance, noting peak yields using sugar beet molasses and *Clostridium* at a specific organic loading rate (OLR).

Skills

Computational & Modeling: MATLAB, Aspen HYSYS

Analytical Techniques: TGA, SEM-EDS, XRD